

Behavior-based scenario discovery using time series clustering

Steinmann, P., Auping, W. L., & Kwakkel, J. H.

This is a "Post-Print" accepted manuscript, which has been Published in "Technological Forecasting and Social Change"

This version is distributed under a non-commercial no derivatives Creative Commons  (CC-BY-NC-ND) user license, which permits use, distribution, and reproduction in any medium, provided the original work is properly cited and not used for commercial purposes. Further, the restriction applies that if you remix, transform, or build upon the material, you may not distribute the modified material.

Please cite this publication as follows:

Steinmann, P., Auping, W. L., & Kwakkel, J. H. (2020). Behavior-based scenario discovery using time series clustering. Technological Forecasting and Social Change, 156, [120052]. https://doi.org/10.1016/j.techfore.2020.120052

You can download the published version at:

https://doi.org/10.1016/j.techfore.2020.120052

Behavior-based Scenario Discovery using time series clustering

Patrick Steinmann[a], Willem L. Auping[b], Jan H. Kwakkel[b]

[a] *Biometris, Wageningen University & Research, Wageningen, The Netherlands*
[b] *Faculty of Technology, Policy & Management, Delft University of Technology, Delft, The Netherlands*

Abstract

Scenario Discovery is a widely used method in model-based decision support for identifying common input space properties across ensembles of exploratory model runs. For model runs with behavior over time, these properties are identified by reducing each run to a single value, which obscures potentially decision-relevant dynamics. We address the problem of considering dynamics in Scenario Discovery by applying time series clustering to the ensemble of model runs, and then finding the common input properties for each cluster. This separates the input space into multiple scenarios, each corresponding to a distinct model dynamic. Policy interventions can be targeted at different scenarios by analyzing overlap of these subspaces. Our work expands Scenario Discovery by improving consideration of system behavior over time, which is highly relevant for the management of complex nonlinear systems such as ecosystems or technical infrastructure.

Keywords: scenarios, scenario discovery, exploratory modelling, time series clustering, deep uncertainty, policy analysis

1. Introduction

Many significant societal challenges are wicked problems [1]. A significant hurdle in understanding these challenges is the deep [2] or Knightian [3] uncertainty surrounding them. That is, the various parties involved do not know or cannot agree on the key mechanisms of the system, associated probability dis-

tributions, and which outcomes are of interest [2]. This is especially relevant for socio-environmental and -technical systems which provide essential ecosystem services, but face increasingly uncertain and volatile futures [4].

Because of the intrinsic complexity of many wicked problems, model-based analyses are useful for supporting decision making and planning [5, 6]. Human cognition struggles in the face of complexity due to the misapprehension of feedbacks, accumulation, time delays, and emergence [7]. Simulation models can be used to augment human reasoning by assessing the consequences of multiple interacting (non-linear) processes [8]. However, this leaves unresolved the problem of uncertainty. [9] suggested to use computational experimentation across many alternative realizations of the various irreducible uncertain factors to systematically map out the consequences of the uncertainties. Scenario Discovery [10, 11] was suggested as a way of analyzing the results of these computational experiments in order to extract decision-relevant information from them.

Scenario Discovery is by now an established model-based technique for scenario development [see e.g. 12, 13, 14, 15, 16, 17, 18, 19, 20, 21]. It also forms the analytical heart of various approaches for model-based decision support under deep uncertainty [22, 23, 24, 25, 26].

The term "scenario" has various definitions and meanings [27]. Scenario Discovery traces its roots to the Intuitive Logics [28] school of scenario planning [10]. Analogous to Intuitive Logics, Scenario Discovery starts with the identification of uncertain factors. Next, Intuitive Logics groups these factors into clusters or mega trends and identifies the most uncertain, most impactful groups which will make up the scenario logic. For an example of how the narrative-first approach of Intuitive Logics can be used to reduce uncertainty, see [29]. In contrast, Scenario Discovery uses computational experimentation with models to explore the implications of all the uncertain factors jointly, then the outputs of the experiments are classified as being of interest or not, before trying to find orthogonal subspaces in the uncertainty space which are predictive of the experiments of interests. In Scenario Discovery, this subspace is called a 'scenario', although sometimes the individual experiments are also confusingly called sce-

narios. Abstractly put, however, this is only a difference in where and how the dimensionality of the uncertainty space is reduced into a smaller interpretable set. Scenario Discovery thus reduces a problem's complexity by projecting the entire uncertainty space into a few salient dimensions, thus rendering cognitive benefits for decision makers [10] and enabling clearer analysis of decision options and trade-offs [25]. This has also been described as model salience [30].

Central to Scenario Discovery is the classification of experiments as being of interest (or "decision-relevant") or not [31, 32]. In virtually all applications of Scenario Discovery, this is done by comparing an outcome of interest for each experiment with an external threshold. Experiments are considered of interest if the outcome meets (or fails to meet, as may be the case) this threshold. That is, the dynamics of the outcome of interest over time are ignored, with some even arguing that they are not relevant [33]. Evaluating system states against such a static criterion is easy to conceptualize and communicate to stakeholders.

However, a problem's dynamics may be crucial for management and policy [34, 35]. That is, decision makers might very well care about the dynamics over time of the outcome of interest. Analytically, dynamics are also relevant. Often, in particular in case of non-linear models, different temporal dynamics originate from different regions of the model input space. Lumping these different dynamics together through a static criterion obfuscates the different origins of the types of dynamics. When using Scenario Discovery for designing strategies, as is common in Robust Decision Making, different temporal dynamics plausibly constitute different vulnerabilities. An inability to separate these vulnerabilities because of the use of a static criterion can result in an inability to succinctly describe the subspace(s) from which the experiments of interest originate. This will in turn hamper an analyst's ability to design robust strategies.

In short, aggregate statistics of time series may be misleading [36], or such time series may be equifinal but dynamically distinct [37], impeding the discovery of decision-relevant scenarios from them at a single point in time. This paper addresses the problem of accounting for dynamics over time within Scenario Discovery.

We present *behavior-based Scenario Discovery* as a method to address the problem of accounting for dynamics over time in Scenario Discovery. Rather than identifying a single subset of decision-relevant simulation experiments by evaluating an outcome of interest at a particular point in time against an external criterion, we apply time series clustering to the entire ensemble of experiment outputs, partitioning it into multiple behaviorally distinct subsets. Next, we can identify the subspaces from which these behaviorally distinct subsets originate. Each of these represents a distinct alternative future (or scenario), and can then be used to support scenario-based decision making [28, 21, 38, 16, 29], or for computational decision support under deep uncertainty [22, 23, 24, 26]. This approach builds on previous efforts to combine clustering and Scenario Discovery [16]. We extend this work with a more effective clustering algorithm, and consideration of the full output ensemble.

In Section 2, we review the Scenario Discovery literature in more detail. In Section 3, we propose an integration of time series clustering into Scenario Discovery to account for dynamics over time. In Section 4, we illustrate behavior-based Scenario Discovery with a case study on the impact of the shale gas revolution on future oil prices. In Section 5, we discuss the method's contributions to both Scenario Discovery in particular and model-based decision making in general, and give an outlook on future research. In section 6, we summarize our main conclusions and contributions.

2. Literature Review

[39] and [22] are the first reported applications of what is now called Scenario Discovery. [31] introduced the name Scenario Discovery, while exploring the potential of various rule induction algorithms for analysing large ensembles of computational experiments with simulation models generated using Exploratory Modelling [9, 40]. Scenario Discovery as it is being used today was put forward by [10].

In Table 1, we summarise the general steps of Scenario Discovery. Sce-

nario Discovery starts with large scale computational experimentation with one or more simulation models. These experiments are designed to systematically cover the space spanned by the many uncertain factors associated with a model. Next, the outputs of the model for each experiment are classified as being of interest or not. Third, a rule induction algorithm, typically the Patient Rule Induction Method (PRIM) [41], is used to identify orthogonal subspaces within the uncertainty space that have a high concentration of experiments of interest. Finally, the intervals for the various uncertain factors which jointly characterize the orthogonal subspace are interpreted and communicated, possibly in the form of narratives [see e.g., 42], but sometimes also directly with evident success [43]. More in depth descriptions can be found in [10], and [11].

Scenario Discovery has been used in the context of climate adaptation [e.g. 12, 49], climate mitigation [e.g. 42, 15, 20, 13, 14, 17, 21], water resources management [e.g. 50, 51], transport and logistics [e.g. 19], material scarcity [e.g. 16], and national security [e.g. 52]. Scenario Discovery is also at the heart of various robust decision making approaches [22, 23], and adaptation pathways [53, 54].

Scenario Discovery conventionally uses a binary classification to identify model outputs as being decision-relevant or not [10]. In such cases, only a single scenario region of the input parameter space can be identified. This approach can limit the usefulness of Scenario Discovery, as scenario-based planning benefits from having multiple comparable and internally consistent scenarios with which to evaluate possible future developments and interventions [55, 29].

The Scenario Discovery literature contains only a small number of studies that have identified multiple distinct scenarios using Scenario Discovery. Both [17] and [46] do not explicitly specify outcomes of interest, instead choosing to split the model output space into multiple regions representing different Shared Socioeconomic Pathways [56]. For the model outputs in each region, Scenario Discovery is performed in turn to identify the underlying drivers. In a study on economic growth, energy consumption, and carbon emissions, [14] partitions the input parameter space by clustering model outputs along multiple decision-relevant output dimensions, and then using another rule induction algorithm,

Table 1: Steps for conventional Scenario Discovery

Step	Action	Comments
1. Generation	Conduct simulation experiments on a system model by sampling from the input parameter space, passing these inputs to the model and recording the generated outputs.	The simulation model acts as a black-box generator function [22], making this step agnostic to the underlying simulation paradigm (differential equation system, agent-based model, discrete event simulation, etc.) Input sampling should be uniform to ensure unbiased and complete coverage of the parameter space. Sampling density can vary, and should be chosen carefully based on computing infrastructure and model complexity [33, 44, 45].
2. Identification	Reduce each model output to a single value. Evaluate this value against an external criterion. Model outputs fulfilling the criterion are identified as being "of interest".	"Model output" refers to all data captured for a single simulation experiment, which may include multiple variables. Analysts must then identify which variable is especially descriptive of the system's overall condition, and how to reduce it to a single value. Depending on the context, this might be an end value, mean, amplitude, or other statistical simplification. The external criterion is a proxy or direct specification of stakeholder goals. It usually takes the form of a threshold value [42], but could also be a range [46].
3. Rule Induction	Find the subspace (or region) in the input parameter space where the inputs generating the outputs of interest lie, by inducing its parameter rules.	Various rule induction algorithms exist [31, 11, 32, 46, 47]. They generally function by drawing a bounding box around the entire input space, and then iteratively restricting the size of this box along one or more axes of the space. These restrictions are referred to as rules, since they prescribe which values a variable can take. The restriction process is governed by three key attributes of the induced subspace - coverage, density, and interpretability [22, 47]. Coverage refers to the ratio of decision-relevant over total inputs contained in the subspace, and should be maximized. Density captures the ratio of decision-relevant over decision-irrelevant inputs in the subspace, and should also be maximized. Interpretability describes the number of parameter space axes along which the box is restricted in size, and should be minimized. Rule induction is generally conducted once, as there is only a single subset of decision-relevant model outputs. However, these outputs may stem from multiple regions of the input space, requiring multiple passes [10, 46]. Repeated rule induction for the same subset of inputs can also improve the quality of the box describing their subspace [48].

Classification and Regression Trees (CART) [57, 31], to identify the input parameters most useful for distinguishing clusters. While analyzing conditions under which the European Emission Trading System would fail to meet its stated objectives, [15] identify three different subspaces using Scenario Discovery. Based on an in depth explanation of why under each subspace, the emission trading system would fail, they conclude that these subspaces represent clearly distinct vulnerability scenarios. It is noteworthy that none of these aforementioned studies analyze relations between the identified scenarios, such as overlap or adjacency. In this work, we show that analysis of scenario overlap is both possible and can yield novel and policy-relevant insights.

3. Behavior-Based Scenario Discovery

Behavior-based Scenario Discovery is a variation of conventional Scenario Discovery, with the goal of deriving decision-relevant future scenarios from model behaviors. Behavior-based Scenario Discovery primarily entails a modification of step 2 (see Table 1). Instead of imposing a binary classification on a set of computational experiments based on model outputs, behavior-based Scenario Discovery applies time series clustering to the model outputs. This enables the separation of distinct model behaviors. In step 3, rule induction can be applied to each cluster of behavior which is deemed to be of interest given the purpose of the study.

Time series clustering aims at grouping a set of time series into two or more subsets with high similarity within each subset, and low similarity across the subsets [58, 59]. [60] identifies three main approaches to quantifying time series similarity: feature-based, data-based, and model-based methods. Feature-based methods identify salient properties of time series, such as amplitude or number of peaks [see e.g. 61]. Similarity between time series is then a function of these features. Data-based methods match individual data points between time series, and determine the points' similarity using various distance definitions [see e.g., 62, 63]. Model-based methods replace each time series with a mathemat-

ical model of itself, such as a linear regression or Markov Chain Monte Carlo distribution, and then determine similarity based on the parameters of those models [see e.g. 64].

Based on the determined similarities, the set of time series can be grouped into a number of internally similar clusters. Clustering methods can be hard, soft, or hierarchical [59]. Hard clustering means every time series is included in exactly one cluster, while in soft clustering, time series are members of multiple clusters, to varying degrees [58]. Hierarchical clustering also assigns every time series to multiple clusters, but at different levels of aggregation (see e.g., [65, 14]).

The starting point of behavior-based Scenario Discovery is a verified and validated system model, together with an input parameter space capturing the model's aleatory and epistemic uncertainties [66]. The first step is identical to that of conventional Scenario Discovery - a number of simulation experiments are performed on a system model, randomly sampling from the input parameter space to generate outputs. In the second step, time-series clustering is applied to find common macro-level behaviors in the ensemble of output time series. In step 3, rule induction is performed for each time series cluster in turn.

The primary benefits of this approach over conventional Scenario Discovery are two-fold:

1. The induced regions in the input parameter space are associated with model behaviors over time, rather than a static snapshot of the model's state.

2. Every output in the ensemble is included in a cluster and mapped to a region in the input parameter space, rather than just a small subset.

In turn, we suggest that the induced regions in the input parameter space, since they are associated with a given type of dynamics over time, will be more clearly separable compared to the existing practice of using a static threshold based approach for identification of experiments which are of interest.

4. Case study: Behavior-based Scenario Discovery for future oil price dynamics

In this Section, we demonstrate behavior-based Scenario Discovery through a case study. To showcase its benefits, it is necessary to analyse a complex system model with both a significant number of uncertain factors, and highly nonlinear dynamics. Therefore, we use a study on climate mitigation policies, developed by [67] using a System Dynamics approach [68]. The model includes both uncertain input parameters as well as uncertainty about model structure in order to capture the ambiguity of the global energy market, and, as will be shown, has highly nonlinear dynamics.

4.1. Generation

We re-used an existing data set created by [67]. The data includes 2000 simulation experiments generated using the Exploratory Modelling workbench [69] under uniform Latin Hypercube sampling [70]. Figure 1 illustrates the diversity of model behaviors that result from different combinations of input parameters. The wide variety of possible behaviors and the non-linearity of the model is apparent.

4.2. Identification of dynamics

We focused our analysis on one particularly interesting variable in the model outputs - future oil price (*Oil Price* [$/$_{BBTU}$]). The ensemble of outputs for this variable, shown in Figure 1, features a variety of dynamics, including oscillations and extreme outliers.

We applied time series clustering with typical settings to the ensemble of model outputs. We chose Complexity-Invariant Distance (CID) [71] as our similarity metric. In [72], we compared a range of time series clustering metrics. We found that CID-based time series clustering was best able to identify tipping points for two simple nonlinear systems, one featuring a saddle-node bifurcation [73], the other a Hopf bifurcation [74]. Given that CID performed best on these two test problems, where we know analytically the correct answer, we

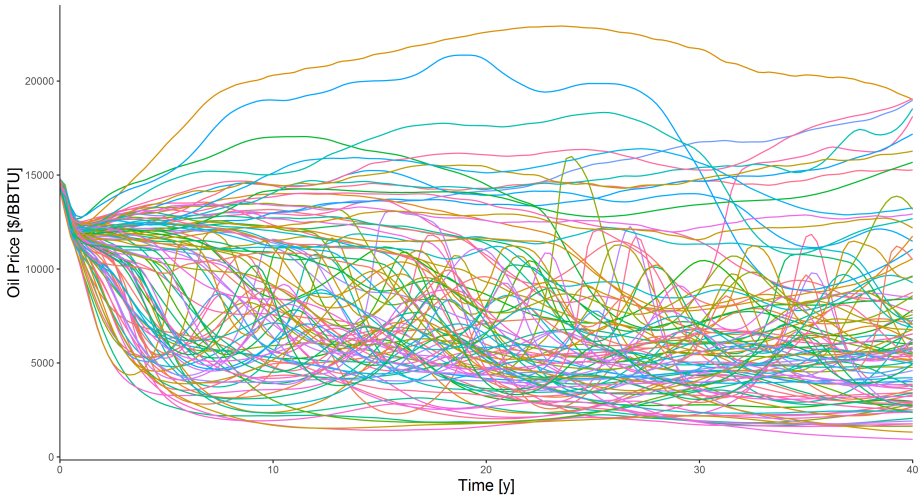


Figure 1: [67] data set shows a variety of nonlinear dynamics. 100 (of 2000) randomly selected outputs for the outcome of interest, each representing a possible alternative future oil price.

expect that CID is likely to perform well on more complex, non-linear dynamical systems as well.

CID is based on Dynamic Time Warping [62], but introduces a correction factor to account for the time series complexity. For every pair of time series Q, C, the complexity-invariant distance between them is calculated as the summed Euclidean distance ED between their points, multiplied by a correction factor CF:

$$CID(Q,C) = ED(Q,C) * CF(Q,C)$$

The correction factor is the quotient of the time series' minimal and maximal complexity estimates:

$$CF(Q,C) = \frac{max(CE(Q), CE(C))}{min(CE(Q), CE(C))}$$

To estimate a time series' complexity, [71] consider its path length:

$$CE(Q) = \sqrt{\sum_{i=1}^{n-1} (q_i - q_{i+1})^2}$$

Based on internal validation [75] of the solutions for different cluster counts, we selected a cluster count of $k = 6$. Hard clustering (i.e. each time series is assigned to a single cluster) was performed using the TSclust package [76], which implements CID clustering for the R computing environment [77]. More recently, CID-based time series clustering has also been implemented in the Python-based EMA workbench [69]. Figure 2 shows the six identified clusters, each representing a set of distinct future dynamics. Based on visual inspection it is apparent that CID-based time series clustering does a reasonably good job of separating different dynamics. For example, if we compare clusters #1 and #2 we see clearly different dynamics. Cluster #1 contains runs that show an early drop and then stay low, with some runs showing small oscillations. In contrast, cluster #2 contains runs where the price drops slightly, but then quickly recovers and stay high for the remainder of the simulation. For some of the other clusters, the differences are less striking. For example, both clusters #4 and #5 seem to contain many oscillating dynamics in the middle of the output range.

4.3. Rule Induction

We performed rule induction for each cluster using the Patient Rule Induction Method (PRIM) [41] as implemented in the Exploratory Modelling workbench [69]. In Table 2, we present the regions of the input parameter space for each cluster. PRIM identifies a region by restricting input parameter values with rules. We give the rules for each cluster along the five most commonly predictive dimensions, along with the coverage and density of each cluster's orthogonal box (see Table 1 for a formal definition of these concepts).

For each of the six clusters, at least one of the five most predictive dimensions was not predictive at all, indicating that the parameters are only predictive for certain behavior modes, and only over a part of their uncertainty range. This underlines the inherent complexity and richness of the underlying model, and has both positive and negative aspects. While it may make it easier to target policy interventions on particular behaviors, it also may be harder to identify

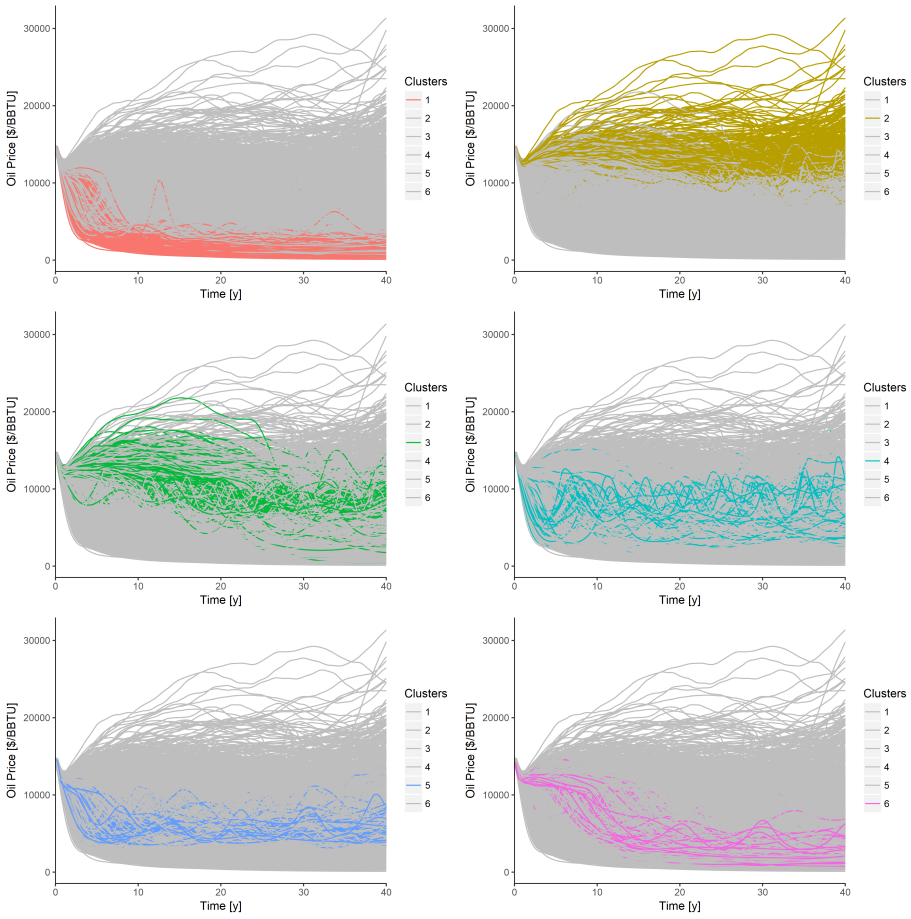


Figure 2: Six subsets of model dynamics, clustered using Complexity-Invariant Distance.

Table 2: Induced rules for most commonly predictive input parameters, per cluster

Cluster	**Rules**					**Box Statistics**	
	Switch price/supply dominance *{1, 2}*	Initial unit costs oil *[1000, 8000]*	Effect of supply shortage on GDP growth *[-0.3, 0]*	Switch legal emissions cap *{0, 1, 2, 3}*	Average throughput time stocks *[0.05, 0.2]*	Coverage *[0, 1]*	Density *[0, 1]*
1	1	[1002, 4945]	[-0.299, -0.088]	{0, 1, 2, 3}	[0.057, 0.2]	0.658	0.649
2	2	[1000, 8000]	[-0.27, 0]	0	[0.05, 0.2]	0.613	0.807
3	2	[1000, 8000]	[-0.22, 0]	{1, 2, 3}	[0.05, 0.2]	0.594	0.449
4	1	[3368, 7997]	[-0.3, -9.6e-05]	{0, 1, 2, 3}	[0.05, 0.11]	0.615	0.548
5	1	[2293, 7753]	[-0.299, -0.015]	{0, 1, 2, 3}	[0.084, 0.2]	0.724	0.766
6	2	[1002, 5819]	[-0.282, -0.028]	{1, 2, 3}	[0.05, 0.2]	0.655	0.555

globally representative signposts and triggers for adaptive policy design. As an example, oil price will remain constant or increase (cluster #2) without a global emissions cap of some kind, due to unbridled demand and consumption. This indicates that an emissions cap may be an effective way to reduce oil prices long-term. However, there are also some cases in which prices stabilize without a cap, or exhibit undesirable behavior despite the presence of a cap. Similarly, cluster #5, which exhibits rapidly fluctuating oil prices, has a low average throughput time for stocks in the model. This mirrors the bullwhip effect sometimes found in stock-flow models or games such as the Beer Game [78], and would indicate that enforcing slower throughput (e.g. by restricting trading) could reduce price fluctuations. However, there are also model runs with little price fluctuation and low throughput time, as well as large fluctuation despite high throughput time.

The induced rules for the *Switch legal emissions cap* variable also show how Scenario Discovery may be useful for model validation - while the categorical input is specified with four levels $\{0, 1, 2, 3\}$, only two can reliably be distinguished: 0 and $\{1, 2, 3\}$. This indicates that this part of the model structure may be unnecessarily specific - or conversely, that the emissions cap levels are not graduated well enough.

4.4. Analysis of Scenario Regions

One advantage of behavior-based Scenario Discovery is that policy interventions can be targeted at specific types of dynamics in light of the identified conditions under which they occur (i.e, the identified region in the input parame-

ter space). To ensure that such interventions only affect the intended dynamics, the respective input parameter regions of each scenario must be well separated from other scenarios, with little to no overlap. Based on the explicit boundaries for each subspace region (or "box") induced by PRIM, we analysed the overlaps between them. As discussed in Section 2, this analysis, although potentially insightful from a multi-scenario perspective, had never previously been performed.

In Table 3, we give both inherent and relational properties for each induced box. Specifically, we list how many time series were attributed to the corresponding cluster, and how many of their inputs ended up inside the corresponding induced input region (true positives). We also show how many members of other clusters were included in the region (false positives).

Table 3: Members of input parameter regions

Region	Total in targeted cluster	Total in induced region	Included members of cluster 1	Included members of cluster 2	Included members of cluster 3	Included members of cluster 4	Included members of cluster 5	Included members of cluster 6
1	386	391	254	1	1	41	69	25
2	287	218	1	176	37	3	0	1
3	426	564	16	61	253	17	13	204
4	234	263	36	11	21	144	49	2
5	308	291	164	16	65	73	223	38
6	359	423	34	14	106	15	19	235

It is apparent that while each region predominantly includes the members of its targeted cluster, there is a significant number of false positives. We believe this is due to the orthogonality of the input space regions induced by PRIM, which has been criticized before [79, 80].

For policy design, overlap between the regions is a crucial point of analysis. To ensure an intervention targeting a specific scenario does not have unintended consequences for other input space regions, the regions must be clearly separated in the multidimensional input parameter space. In Table 4, we quantify the overlap between the six scenario regions by determining how many experiments they share.

It is apparent that the regions are separable - only 4 of 15 pairs overlap. The overlaps for box pairs #1 and #5, as well as #3 and #6, are noteworthy.

Table 4: Shared experiments (overlap) for induced scenario regions

Boxes	1	2	3	4	5	6
1	391	0	0	65	208	0
2		218	0	0	0	0
3			564	0	0	324
4				263	100	0
5					291	0
6						423

The clustering solution in Figure 2 shows that the pairs have similar behavior, and Table 2 indicates they share structural uncertainty parameters. It is therefore not surprising that they overlap to some degree. However, it may still be interesting to distinguish them for scenario purposes, especially #3 and #6, as the latter shows a more severe oil price drop than the former, which could have significant policy implications [81].

5. Discussion

5.1. Case study results

In our case study, we chose to use Complex Invariant Distance in combination with hard clustering as the method for performing time series clustering. The central claims of this paper do not rest on this particular choice. It is conceivable that for other models, e.g. models characterized by more discrete temporal variations, other time series clustering algorithms are more suitable. Still, as evidenced by Figure 2 and the more in depth analysis of the separability of the clusters in the model input space, CID did perform quite well. As such, we suggest that CID is a promising first choice method for time series clustering in the context of behavior-based Scenario Discovery.

We combined CID with agglomerative clustering, which results in a hard assignment of individual time series to a specific cluster. The analysis of the overlap between the induced regions in Table 4 as well as the visual inspection suggests that a soft clustering might have been more appropriate. Only cluster 2 is perfectly separable in the model input space. While the other clusters are harder to separate both in the input space as well as having some apparent overlap based on visual inspection. It is unclear however how to embed

a soft cluster assignment within Scenario Discovery, given its reliance on rule induction, which requires a binary classification.

5.2. Contribution to Scenario Discovery

Scenario Discovery is a computational approach for identifying consistent, decision-relevant futures [10]. For the management of nonlinear systems, understanding the dynamics and behaviors of these futures is crucial [82]. However, conventional Scenario Discovery analyzes model outputs at a single point in time, ignoring these dynamics. With behavior-based Scenario Discovery, we present a method of deriving decision-relevant scenarios from these dynamics. Our approach builds on previous work regarding static vs. dynamic (or "transient") scenarios [83], identifying common patterns in complex model outputs [16], and inducing multiple decision-relevant input space regions [17, 46, 15, 14]. We combine these lines of research into a novel enhancement of Scenario Discovery, capable of identifying and characterizing behaviorally distinct futures. We see our work as a contribution to the ongoing discussion and development of Scenario Discovery in this journal [84, 10, 40, 16, 48] and elsewhere.

5.3. Contribution to model-based decision support

The presented approach for applying Scenario Discovery to ensembles of dynamic model runs is of particular interest in the context of an ongoing discussion on the use of Scenario Discovery for developing global change scenarios. The existing SSP/RCP framework is based on a story line and simulate approach [85]. First, internally consistent narratives are developed by groups of experts. Next, these narratives are translated into sets of inputs for specific integrated assessment models for quantitatively exploring specific Shared Socio Economic Pathways [56], or in combination with Reference Concentration Pathways.

[17] argued that the *a priori* specification of challenges to mitigation and adaptation is problematic, instead favouring *a posteriori* identification assisted by Scenario Discovery. One advantage of such a simulate and storyline approach [42] is that it offer better guarantees of properly bounding the space of

possible futures. [46] methodologically developed this idea further. [21] demonstrated the feasibility of this approach by systematically sampling all possible combinations of drivers from the SSP dimensions, evaluating them using the integrated assessment model GCAM. They empirically confirmed the theoretical point raised by [17] - similarly looking challenges to adaptation and mitigation can arise from quite disparate combinations of driving forces. The approach presented in this paper offers a further methodological extension allowing for the explicit consideration of temporal dynamics.

5.4. Future Research

We see three main avenues for future work on behavior-based Scenario Discovery. Firstly, we hope to see our analytical method tested with other dynamic models to validate its general applicability. In particular, it may be interesting to consider stochastic models, such as agent-based or discrete event models, as it remains to been how sensitive our proposed approach is to noisy time series outputs. To this end, we have made our code publicly available both on GitHub (https://github.com/steipatr/BBSD-Public), and in the Exploratory Modelling Workbench [69].

Secondly, the method presented in this paper uses univariate clustering to group the model outputs. However, multivariate clustering—grouping simulation experiments based on the similarities of multiple time-dependent model variables at once—may more reliably distinguish model dynamics by encompassing a wider range of system performance markers. It might also help integrate differing stakeholder opinions about which system variables should form the basis for deliberation—a core challenge in decision-making under deep uncertainty [2, 84, 26]. A similar extension from univariate to multivariate Scenario Discovery, motivated by the same considerations, has already been dpresented by [14] for analysis at a single point in time.

Thirdly, we found that scenario regions overlap in some cases. We believe this is due to inherent orthogonality limitations of the used rule induction algorithm, which have been discussed elsewhere [79, 80]. Developing non-orthogonal rule

induction methods, possibly based on genetic algorithms [47], may reduce region overlap and improve separability, allowing more precise identification of the root causes of specific model behaviors.

6. Conclusions

We have presented behavior-based Scenario Discovery. Where the established practice of Scenario Discovery virtually always focuses on the outcomes of a model at a particular point in time, behavior-based Scenario Discovery enables the analysis of dynamics over time instead. This is relevant for several reasons. In the context of designing robust strategies through Robust Decision Making, different undesirable dynamics over time constitute different kinds of vulnerabilities. By analyzing the model outcomes at a particular point in time, analysts will be unable to differentiate these dynamically distinct vulnerabilities. Second, different types of dynamics over time often originate from distinct regions of the uncertainty space. Explicitly considering dynamics over time within Scenario Discovery will improve the partitioning of the uncertainty space that can be produced by the existing rule induction algorithms used within Scenario Discovery, such as PRIM.

Behavior-based Scenario Discovery replaces the existing threshold-based binary classification of computational experiments with a time series clustering step. Next, rule induction can be performed for each cluster of computational experiments. As demonstrated in our case study, behaviorally distinct future scenarios can be identified by clustering an ensemble of time series model outputs, and characterized by inducing and analyzing the subspaces in the uncertainty space from which they originate. These scenarios based on dynamics can then be used for a variety of model-based decision support purposes, including uncertainty exploration, adaptive policy design, and strategic planning.

We focused in this paper on the clustering of time series as a precursor to rule induction within Scenario Discovery. We suggest that our arguments in favor of this multinomial approach in case of temporal dynamics apply more

broadly. For example, instead of imposing a binary classification on the results of a spatially explicit model, it might be more useful to first cluster based on spatial patterns, and then use Scenario Discovery on these clusters. In particular in the context of land use change models, as used for e.g., climate adaptation, such an approach might be fruitful for identifying the drivers behind distinct land use patterns.

References

[1] H. W. J. Rittel, M. M. Webber, Dilemmas in a general theory of planning, Policy Sciences 4 (2) (1973) 155–169. doi:10.1007/bf01405730.

[2] R. J. Lempert, S. W. Popper, S. C. Bankes, Shaping the next one hundred years: new methods for quantitative, long-term policy analysis, Report, RAND Corporation (2003).
URL https://www.rand.org/pubs/monograph_reports/MR1626.html

[3] F. H. Knight, Risk, uncertainty and profit, Houghton Mifflin Company, Boston, 1921.

[4] D. Helbing, Globally networked risks and how to respond, Nature 497 (2013) 51, perspective. doi:10.1038/nature12047.

[5] J. Kwakkel, W. Walker, M. Haasnoot, Coping with the wickedness of public policy problems: Approaches for decision making under deep uncertainty, Journal of Water Resources Planning and Management 142 (3). doi:10.1061/(ASCE)WR.1943-5452.0000626.

[6] G. Holtz, F. Alkemade, F. de Haan, J. Khler, E. Trutnevyte, T. Luthe, J. Halbe, G. Papachristos, E. Chappin, J. Kwakkel, R. S., Prospects of modelling societal transitions: Position paper of an emerging community, Environmental Innovation and Societal Transitions 17 (2015) 41–58. doi:10.1016/j.eist.2015.05.006.

[7] J. Sterman, Learning in and about complex systems, System Dynamics Review 10 (23) (1994) 291–330. doi:10.1002/sdr.4260100214.

[8] J. Sterman, All models are wrong: reflections on becoming a systems scientist, System Dynamics Review 18 (4) (2002) 501–531. doi:10.1002/sdr.261.

[9] S. Bankes, Exploratory modeling for policy analysis, Operations Research 41 (3) (1993) 435–449. doi:10.1287/opre.41.3.435.

[10] B. P. Bryant, R. J. Lempert, Thinking inside the box: A participatory, computer-assisted approach to scenario discovery, Technological Forecasting and Social Change 77 (1) (2010) 34–49. doi:10.1016/j.techfore.2009.08.002.

[11] J. H. Kwakkel, M. Jaxa-Rozen, Improving scenario discovery for handling heterogeneous uncertainties and multinomial classified outcomes, Environmental Modelling & Software 79 (2016) 311–321. doi:10.1016/j.envsoft.2015.11.020.

[12] R. Lempert, D. Groves, Identifying and evaluating robust adaptive policy responses to climate change for water management agencies in the american west, Technological Forecasting and Social Change 77 (6) (2010) 960–974. doi:10.1016/j.techfore.2010.04.007.

[13] H. C. McJeon, L. Clarke, P. Kyle, M. Wise, A. Hackbarth, B. P. Bryant, R. J. Lempert, Technology interactions among low-carbon energy technologies: What can we learn from a large number of scenarios?, Energy Economics 33 (4) (2011) 619 – 631, special Issue on The Economics of Technologies to Combat Global Warming. doi:10.1016/j.eneco.2010.10.007.

[14] M. Gerst, P. Wang, M. Borsuk, Discovering plausible energy and economic futures under global change using multidimensional scenario discovery, Environmental Modelling & Software 44 (2013) 76–86. doi:10.1016/j.envsoft.2012.09.001.

[15] C. Hamarat, J. Kwakkel, E. Pruyt, Adaptive robust design under deep uncertainty, Technological Forecasting and Social Change 80 (3) (2013) 408–418. doi:10.1016/j.techfore.2012.10.004.

[16] J. H. Kwakkel, W. L. Auping, E. Pruyt, Dynamic scenario discovery under deep uncertainty: The future of copper, Technological Forecasting and Social Change 80 (4) (2013) 789–800. doi:10.1016/j.techfore.2012.09.012.

[17] J. Rozenberg, C. Guivarch, R. Lempert, S. Hallegatte, Building ssps for climate policy analysis: a scenario elicitation methodology to map the space of possible future challenges to mitigation and adaptation, Climatic Change 122 (3) (2014) 509–522. doi:10.1007/s10584-013-0904-3.

[18] A. Parker, S. Srinivasan, R. Lempert, S. Berry, Evaluating simulation-derived scenarios for effective decision support, Technological Forecasting and Social Change 91 (February) (2015) 64–77. doi:10.1016/j.techfore.2014.01.010.

[19] R. A. Halim, J. H. Kwakkel, L. A. Tavasszy, A scenario discovery study of the impact of uncertainties in the global container transport system on european ports, Futures 81 (2016) 148–160. doi:10.1016/j.futures.2015.09.004.

[20] E. A. Moallemi, F. de Haan, J. Kwakkel, L. Aye, Narrative-informed exploratory analysis of energy transition pathways: A case study of india's electricity sector, Energy Policy 110 (2017) 271 – 287. doi:10.1016/j.enpol.2017.08.019.

[21] J. R. Lamontagne, P. M. Reed, R. Link, K. V. Calvin, L. E. Clarke, J. A. Edmonds, Large ensemble analytic framework for consequence-driven discovery of climate change scenarios, Earth's Future 6 (3) (2018) 488–504. doi:10.1002/2017ef000701.

[22] R. J. Lempert, D. G. Groves, S. W. Popper, S. C. Bankes, A general, analytic method for generating robust strategies and narrative scenarios, Management Science 52 (4) (2006) 514–528. doi:10.1287/mnsc.1050.0472.

[23] J. Kasprzyk, S. Nataraj, P. Reed, R. Lempert, Many objective robust decision making for complex environmental systems undergoing change, Environmental Modelling & Software 42 (2013) 55–71. doi:10.1016/j.envsoft.2012.007.

[24] J. D. Herman, P. M. Reed, H. B. Zeff, G. W. Characklis, How should robustness be defined for water systems planning under change?, Journal of Water Resources Planning and Management 141 (10) (2015) 04015012. doi:10.1061/(ASCE)WR.1943-5452.0000509.

[25] C. Helgeson, Structuring decisions under deep uncertainty, Topoi 39 (2018) 257–269. doi:10.1007/s11245-018-9584-y.

[26] J. Kwakkel, M. Haasnoot, Supporting DMDU: A Taxonomy of Approaches and Tools, Springer, 2019, book section 15, pp. 355–374. doi:10.1007/978-3-030-05252-2_15.

[27] M. J. Spaniol, N. J. Rowland, Defining scenario, Futures & Foresight Science 1 (1) (2019) e3. doi:10.1002/ffo2.3.

[28] R. Bradfield, G. Wright, G. Burt, G. Cairns, K. Van Der Heijden, The origins and evolution of scenario techniques in long range business planning, Futures 37 (8) (2005) 795–812. doi:10.1016/j.futures.2005.01.003.

[29] G. Willis, S. Cave, M. Kunc, Strategic workforce planning in healthcare: A multi-methodology approach, European Journal of Operational Research 267 (1) (2018) 250–263. doi:10.1016/j.ejor.2017.11.008.

[30] G. A. K. van Voorn, R. W. Verburg, E. M. Kunseler, J. Vader, P. H. M. Janssen, A checklist for model credibility, salience, and legitimacy to

improve information transfer in environmental policy assessments, Environmental Modelling & Software 83 (2016) 224–236. doi:10.1016/j.envsoft.2016.06.003.

[31] R. J. Lempert, B. P. Bryant, S. C. Bankes, Comparing algorithms for scenario discovery, Report, RAND (2008).
URL www.rand.org/content/dam/rand/pubs/working_papers/2008/RAND_WR557.pdf

[32] S. Dalal, B. Han, R. Lempert, A. Jaycocks, A. Hackbarth, Improving scenario discovery using orthogonal rotations, Environmental Modelling & Software 48 (2013) 49–64. doi:10.1016/j.envsoft.2013.05.013.

[33] P. Davis, S. Bankes, M. Egner, Enhancing strategic planning with massive scenario generation, Report, RAND Corporation (2007).
URL https://www.rand.org/pubs/technical_reports/TR392.html

[34] N. M. Gotts, G. A. K. van Voorn, J. G. Polhill, E. d. Jong, B. Edmonds, G. J. Hofstede, R. Meyer, Agent-based modelling of socio-ecological systems: Models, projects and ontologies, Ecological Complexitydoi:10.1016/j.ecocom.2018.07.007.

[35] G. A. ten Broeke, G. A. K. van Voorn, A. Ligtenberg, J. Molenaar, Resilience through adaptation, PLOS ONE 12 (2) (2017) e0171833. doi:10.1371/journal.pone.0171833.

[36] F. J. Anscombe, Graphs in statistical analysis, The American Statistician 27 (1) (1973) 17–21. doi:10.1080/00031305.1973.10478966.

[37] L. Von Bertalanffy, General System Theory, Braziller, New York, 1968.

[38] M. Kunc, F. A. O'Brien, Exploring the development of a methodology for scenario use: Combining scenario and resource mapping approaches, Technological Forecasting and Social Change 124 (2017) 150–159. doi:10.1016/j.techfore.2017.03.018.

[39] D. Groves, R. Lempert, A new analytic method for finding policy-relevant scenarios, Global Environmental Change 17 (1) (2007) 73–85. doi:10.1016/j.gloenvcha.2006.11.006.

[40] J. H. Kwakkel, E. Pruyt, Exploratory modeling and analysis, an approach for model-based foresight under deep uncertainty, Technological Forecasting and Social Change 80 (3) (2013) 419–431. doi:j.techfore.2012.10.005.

[41] J. H. Friedman, N. I. Fisher, Bump hunting in high-dimensional data, Statistics and Computing 9 (2) (1999) 123–143. doi:10.1023/a:1008894516817.

[42] S. Greeven, O. Kraan, . J. L. Chappin, J. H. Kwakkel, The emergence of climate change mitigation action by society: An agent-based scenario discovery study, Journal of Artificial Societies and Social Simulation 19 (3). doi:10.18564/jasss.3134.

[43] M. Gong, R. Lempert, A. Parker, L. Mayer, J. Fischbach, M. Sisco, Z. Mao, D. Krantz, H. Kunreauther, Testing the scenario hypothesis: An experimental comparison of scenarios and forecasts for decision support in a complex decision environment, Environmental Modelling & Software 91 (2017) 135–144. doi:10.1016/j.envsoft.2017.02.002.

[44] E. Pruyt, T. Islam, On generating and exploring the behavior space of complex models, System Dynamics Review 31 (4) (2015) 220–249. doi:10.1002/sdr.1544.

[45] E. A. Moallemi, S. Elsawah, M. J. Ryan, An agent-monitored framework for the output-oriented design of experiments in exploratory modelling, Simulation Modelling Practice and Theory 89 (2018) 48–63. doi:10.1016/j.simpat.2018.09.008.

[46] C. Guivarch, J. Rozenberg, V. Schweizer, The diversity of socio-economic pathways and co2 emissions scenarios: Insights from the investigation of

a scenarios database, Environmental Modelling & Software 80 (2016) 336–353. doi:10.1016/j.envsoft.2016.03.006.

[47] J. H. Kwakkel, A generalized manyobjective optimization approach for scenario discovery, Futures & Foresight Science 0 (0) (2019) e8. doi:10.1002/ffo2.8.

[48] J. H. Kwakkel, S. C. Cunningham, Improving scenario discovery by bagging random boxes, Technological Forecasting and Social Change 111 (2016) 124–134. doi:10.1016/j.techfore.2016.06.014.

[49] J. Kwakkel, M. Haasnoot, W. Walker, Developing dynamic adaptive policy pathways: A computer-assisted approach for developing adaptive strategies for a deeply uncertain world, Climatic Change 132 (3) (2015) 373–386. doi:10.1007/s10584-014-1210-4.

[50] A. Watson, J. Kasprzyk, Incorporating deeply uncertain factors into the many objective search process, Environmental Modelling & Software 89 (2017) 159–171. doi:10.1016/j.envsoft.2016.12.001.

[51] E. Matrosov, A. Woords, J. Harou, Robust decision making and info-gap decision theory for water resource system planning, Journal of Hydrology 494 (28 June 2013) (2013) 43–58. doi:10.1016/j.jhydrol.2013.03.006.

[52] E. Pruyt, J. Kwakkel, Radicalization under deep uncertainty: a multi-model exploration of activism, extremism and terrorism, System Dynamics Review 30 (1-2) (2014) 1–28. doi:10.1002/sdr.1510.

[53] M. Haasnoot, J. H. Kwakkel, W. E. Walker, J. ter Maat, Dynamic adaptive policy pathways: A method for crafting robust decisions for a deeply uncertain world, Global Environmental Change 23 (2) (2013) 485–498. doi:10.1016/j.gloenvcha.2012.12.006.

[54] J. Kwakkel, M. Haasnoot, W. Walker, Comparing robust decision-making and dynamic adaptive policy pathways for model-based decision support

under deep uncertainty, Environmental Modelling & Software 86 (2016) 168–183. doi:10.1016/j.envsoft.2016.09.017.

[55] P. J. Schoemaker, Multiple scenario development: Its conceptual and behavioral foundation, Strategic management journal 14 (3) (1993) 193–213. doi:10.1002/smj.4250140304.

[56] B. C. O'Neill, E. Kriegler, K. Riahi, K. L. Ebi, S. Hallegatte, T. R. Carter, R. Mathur, D. P. van Vuuren, A new scenario framework for climate change research: the concept of shared socioeconomic pathways, Climatic change 122 (3) (2014) 387–400. doi:10.1007/s10584-013-0905-2.

[57] L. Breiman, J. H. Friedman, R. A. Olshen, C. J. Stone, Classification and Regression Trees, Wadsworth, Monterey, CA, 1984.

[58] J. D. Cryer, K.-S. Chan, Time Series Analysis, 2nd Edition, Springer-Verlag New York, 2008. doi:10.1007/978-0-387-75959-3.

[59] R. H. Shumway, D. S. Stoffer, Time Series Analysis and Its Applications, 4th Edition, Springer Texts in Statistics, Springer International Publishing, 2017. doi:10.1007/978-3-319-52452-8.

[60] T. W. Liao, Clustering of time series dataa survey, Pattern Recognition 38 (11) (2005) 1857–1874. doi:10.1016/j.patcog.2005.01.025.

[61] J. Paparrizos, L. Gravano, k-shape: Efficient and accurate clustering of time series, in: Proceedings of the 2015 ACM SIGMOD International Conference on Management of Data, 2015, pp. 1855–1870. doi:10.1145/2723372.2737793.

[62] D. J. Berndt, J. Clifford, Using dynamic time warping to find patterns in time series., in: AAAI Technical Report WS-94-03, no. 16 in AAAI Reports, Seattle, WA, 1994, pp. 359–370.

[63] E. Keogh, C. A. Ratanamahatana, Exact indexing of dynamic time warping, Knowledge and information systems 7 (3) (2005) 358–386. doi:10.1007/s10115-004-0154-9.

[64] M. Corduas, D. Piccolo, Time series clustering and classification by the autoregressive metric, Computational statistics & data analysis 52 (4) (2008) 1860–1872. doi:10.1016/j.csda.2007.06.001.

[65] P. P. Rodrigues, J. Gama, J. Pedroso, Hierarchical clustering of time-series data streams, IEEE transactions on knowledge and data engineering 20 (5) (2008) 615–627.

[66] F. O. Hoffman, J. S. Hammonds, Propagation of uncertainty in risk assessments: The need to distinguish between uncertainty due to lack of knowledge and uncertainty due to variability, Risk Analysis 14 (5) (1994) 707–712. doi:10.1111/j.1539-6924.1994.tb00281.x.

[67] W. L. Auping, E. Pruyt, S. de Jong, J. H. Kwakkel, The geopolitical impact of the shale revolution: Exploring consequences on energy prices and rentier states, Energy Policy 98 (2016) 390–399. doi:10.1016/j.enpol.2016.08.032.

[68] J. W. Forrester, Industrial Dynamics, Pegasus Communications, Waltham, MA, 1961.

[69] J. H. Kwakkel, The exploratory modeling workbench: An open source toolkit for exploratory modeling, scenario discovery, and (multi-objective) robust decision making, Environmental Modelling & Software 96 (2017) 239–250. doi:10.1016/j.envsoft.2017.06.054.

[70] M. D. McKay, R. J. Beckman, W. J. Conover, Comparison of three methods for selecting values of input variables in the analysis of output from a computer code, Technometrics 21 (2) (1979) 239–245. doi:10.1080/00401706.1979.10489755.

[71] G. E. A. P. A. Batista, E. J. Keogh, O. M. Tataw, V. M. A. de Souza, CID: an efficient complexity-invariant distance for time series, Data Mining and Knowledge Discovery 28 (3) (2014) 634–669. doi:10.1007/s10618-013-0312-3.

[72] P. Steinmann, Behavior-based scenario discovery, Thesis, Delft University of Technology (2018).
URL resolver.tudelft.nl/uuid:cb4fee52-e23f-41d3-9c63-7ca7ec948304

[73] D. Ludwig, D. D. Jones, C. S. Holling, Qualitative analysis of insect outbreak systems: The spruce budworm and forest, Journal of Animal Ecology 47 (1) (1978) 315–332. doi:10.2307/3939.
URL http://www.jstor.org/stable/3939

[74] S. H. Strogatz, Nonlinear dynamics and chaos: with applications to physics, biology, chemistry, and engineering, CRC press, 2018.

[75] O. Arbelaitz, I. Gurrutxaga, J. Muguerza, J. M. Prez, I. Perona, An extensive comparative study of cluster validity indices, Pattern Recognition 46 (1) (2013) 243–256. doi:10.1016/j.patcog.2012.07.021.

[76] P. Montero, J. A. Vilar, Tsclust: An r package for time series clustering, 2014 62 (1) (2014) 43, 1. doi:10.18637/jss.v062.i01.

[77] R Core Team, R: A language for statistical computing (2018).
URL https://www.R-project.org/

[78] D. Meadows, A brief and incomplete history of operational gaming in system dynamics, System Dynamics Review: The Journal of the System Dynamics Society 23 (2-3) (2007) 199–203.
URL 10.1002/sdr.372

[79] J. D. Quinn, P. M. Reed, K. Keller, Direct policy search for robust multi-objective management of deeply uncertain socio-ecological tipping points, Environmental Modelling & Software 92 (2017) 125–141. doi:10.1016/j.envsoft.2017.02.017.

[80] W. L. Auping, Modelling uncertainty: Developing and using simulation models for exploring the consequences of deep uncertainty in complex problems, Thesis, Delft University of Technology (2018). doi:10.4233/uuid:0e0da51a-e2c9-4aa0-80cc-d930b685fc53.

[81] B. Smith, Oil wealth and regime survival in the developing world, 1960–1999, American Journal of Political Science 48 (2) (2004) 232–246. doi:10.1111/j.0092-5853.2004.00067.x.

[82] J. Sterman, Business Dynamics: Systems Thinking and Modelling in a Complex World, McGraw-Hill Education, 2000.

[83] M. Haasnoot, J. Schellekens, J. J. Beersma, H. Middelkoop, J. C. J. Kwadijk, Transient scenarios for robust climate change adaptation illustrated for water management in the netherlands, Environmental Research Letters 10 (10) (2015) 105008. doi:10.1088/1748-9326/10/10/105008.

[84] W. E. Walker, V. A. W. J. Marchau, D. Swanson, Addressing deep uncertainty using adaptive policies: Introduction to section2, Technological Forecasting and Social Change 77 (6) (2010) 917–923. doi:10.1016/j.techfore.2010.04.004.

[85] Y. Garb, S. Pulver, S. D. VanDeveer, Scenarios in society, society in scenarios: toward a social scientific analysis of storyline-driven environmental modeling, Environmental Research Letters 3 (4) (2008) 045015. doi:10.1088/1748-9326/3/4/045015.

Patrick Steinmann received his MSc from Delft University of Technology. He is currently a PhD candidate at Wageningen University & Research, working on quantifying the resilience of socio-environmental systems under deep uncertainty. His main research interest is the nexus of complexity science and long-term decision making.

Willem Auping received his Ph.D. from Delft University of Technology, where he now works as an assistant professor. His research interests are in the area of decision making under deep uncertainty, exploratory modeling and analysis, mineral and metal scarcity, and aging.

Jan Kwakkel works as an associate professor at Delft University of Technology. His research focuses on model based decision support for decision making

under deep uncertainty. He has applied a variety of methods and techniques in different domains. His current focus is climate change in the water sector.